SOFTWARE TEST PLAN

CareConnect Application — Team B | SWEN 670 Summer 2026

Unit Testing • E2E Testing • Coverage Targets • WCAG/508 • IEEE Std 829-1998

Document Version	1.2 — Draft
Standard	IEEE Std 829-1998 Software Test Documentation
Date	June 2026
Prepared By	Alex Yom, Team B Lead
Team	Team B — Comprehensive Testing, WCAG/508 Compliance, VPAT
Members	Alex Yom | Brice Tikum | Prashanth Saseenthar | Lekan Ogunsanya
Milestone	Milestone 2 — Design and Prototyping
Due Date	June 13, 2026

Version History

Version	Date	Description
1.0	June 2, 2026	Initial draft — full document structure, all pre-filled sections, teammate placeholders
1.1	June 2, 2026	Added: defect management, testing risks, test metrics, version history, UAT out-of-scope note
1.2	June 2, 2026	Reassigned Section 8 sprint schedule to Lekan. All sections confirmed assigned.
1.3	June 9, 2026	Added: IEEE 829 document ID, Test Plan Identifier, Features to Be Tested (F-ID table), Test Deliverables, Staffing & Training, Acceptance Criteria, Appendix A execution summary, Glossary, Approval sign-off. Fixed Chime status to operational. Removed collaboration banners. Version updated to 1.3.

Collaboration Color Key

Blue	Brice Tikum — Sections 2, 3, 4: test types/strategy, test cases, coverage targets
Yellow	Prashanth Saseenthar — Section 5: entry/exit criteria, test environment setup
Green	Lekan Ogunsanya — Sections 8, 9: test schedule with sprint detail, deployment/environment context

Gray / ✅ (pre-filled)	Alex Yom — Sections 1, 6, 7, 10, 11, 12: scope, WCAG, roles, defect mgmt, risks, metrics

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0. Test Plan Identifier

This document is assigned the following identifier in accordance with IEEE Std 829-1998, Section 5.1.

Test Plan ID	STP-TEAMB-CARECONNECT-2026-002
Version	1.3 — Draft
Status	Draft — Submitted for milestone review June 13, 2026
Standard	IEEE Std 829-1998: Standard for Software Test Documentation
Project	CareConnect Application — SWEN 670 Software Engineering Capstone, Summer 2026, UMGC
Team Scope	Team B: Comprehensive Testing, WCAG 2.1 AA / Section 508 Compliance, VPAT Production

1. Test Scope & Objectives

✅ **ALEX YOM — Pre-filled. Review for accuracy and update if needed before submission.**

1.1 Purpose

This Software Test Plan defines the testing strategy, scope, test types, coverage targets, roles, and schedule for Team B's contributions to the CareConnect application. It is the operational document guiding all testing activities through Milestones 3 and 4, and is the companion document to the Technical Design Document.

1.2 In Scope

- Unit testing — Flutter/Dart frontend and Spring Boot Java backend
- Widget testing — Flutter UI components including those delivered by SWEN 661
- Integration/E2E testing — critical user workflows: Shift Scheduling, EVV, Authentication, Messaging
- Code coverage measurement and enforcement — tiered thresholds per module category
- WCAG 2.1 AA accessibility testing — automated (axe-core) and manual
- Section 508 compliance testing
- Voice/AI persona accessibility testing — supplemental criteria (WCAG gap)
- VPAT production — ITI VPAT 2.4 format

1.3 Out of Scope

- Feature development — owned by Teams A, C, D, E
- Backend infrastructure deployment — owned by Team A
- Performance/load testing — not required for this course
- Security penetration testing — outside Team B mandate
- User Acceptance Testing (UAT) — deferred; out of scope for Milestones 2 and 3

1.4 References

The following documents are referenced throughout this Software Test Plan:

Reference	Purpose
Team B Software Requirements Specification (SRS)	Defines functional and non-functional requirements to be tested
Team B Technical Design Document (TDD)	Defines testing architecture and tooling decisions this plan implements
Team B Project Plan	Defines milestone deliverables, schedule, roles, risks
Prior Cohort Documents (Spring 2026)	Benchmark for document structure and section coverage
IEEE Std 829-1998	Standard for Software Test Documentation — structural reference for this document
WCAG 2.1 Guidelines	Accessibility criteria for Section 6 test approach
ITI VPAT 2.4 Template	Format for Voluntary Product Accessibility Template deliverable

1.5 Definitions, Acronyms, and Abbreviations

For complete definitions see Section 15 (Glossary). Key terms: STP — Software Test Plan; TDD — Technical Design Document; EVV — Electronic Visit Verification; VPAT — Voluntary Product Accessibility Template; WCAG — Web Content Accessibility Guidelines; CI/CD — Continuous Integration/Continuous Deployment; E2E — End-to-end testing; lcov — Line coverage report format; JaCoCo — Java code coverage tool.

1.6 Test Objectives

The following objectives define what Team B must achieve across Milestones 2, 3, and 4:

- Achieve tiered code coverage: 100% transactional modules, 90% visual/UI, 95% all others
- Validate all critical user workflows pass E2E tests before each milestone submission
- Identify and document all WCAG 2.1 AA violations with remediation owners
- Produce a complete ITI VPAT 2.4 conformance statement by Milestone 4
- Ensure all 661 UI components pass accessibility acceptance criteria before integration

2. Features to Be Tested

The following table identifies all testable features, modules, and system areas within Team B's assigned scope. Each feature is assigned a Feature ID (F-ID), mapped to its corresponding SRS requirement reference, and classified by priority. This table serves as the bridge between the SRS and the test cases defined in Section 4.

F-ID	Feature / Module	Priority	Requirement Reference	Expected Outcome
F-001	Unit Testing Framework — Flutter/Dart	**High**	TR-001, TR-002	466 test files pass; lcov.info generated
F-002	Unit Testing Framework — Spring Boot	**High**	TR-001, TR-002	mvn test BUILD SUCCESS; JaCoCo ≥89%

F-003	Coverage Gate Enforcement — CI	**High**	TR-003, TR-004	Pipeline blocks merge if thresholds unmet
F-004	Shift Scheduling module — unit coverage	**Critical**	TR-001, FR-SS-*	Coverage from 10.1% → 100% (M3)
F-005	EVV module — unit coverage	**Critical**	TR-001, FR-EVV-*	Coverage from 29.5% → 100% (M3)
F-006	Authentication / JWT — unit coverage	**High**	TR-001, FR-AUTH-*	Coverage ≥95% (M3)
F-007	E2E — Full caregiver shift workflow	**High**	ET-001, ET-002	All steps succeed end-to-end
F-008	E2E — EVV visit workflow	**High**	ET-001, ET-002	Check-in through report generation passes
F-009	E2E — Admin workflow	**Medium**	ET-001	Schedules + report generation succeed
F-010	E2E — Messaging (Chime SDK)	**High**	ET-001, ET-003	All 10 Chime E2E tests pass (confirmed M2)
F-011	WCAG 2.1 AA — all screens	**High**	AC-001 – AC-009	No critical or serious axe-core violations
F-012	Section 508 compliance	**High**	AC-001 – AC-009	Keyboard nav, screen reader, contrast pass
F-013	Voice/AI persona supplemental criteria	**Medium**	VP-001	5 supplemental criteria validated (M3/M4)
F-014	VPAT production — ITI 2.4	**High**	VP-001 – VP-004	Interim VPAT M3; final VPAT M4
F-015	661 UI component accessibility gate	**Medium**	AC-001 – AC-009	Each component passes axe-core before merge
F-016	Regression suite — post-Monday-merge	**High**	ET-003	All existing tests pass after integration

2.1 Features Not to Be Tested

The following features are outside Team B's testing scope for Milestone 2. They are documented here to prevent confusion and to clarify ownership boundaries:

Feature / Area	Reason	Owner
Feature development (new code)	Team B tests features; we do not write feature code	Teams A, C, D, E
AWS infrastructure deployment	DevOps scope — not testing scope	Team A
HHAeXchange API internal testing	Third-party system — outside course scope	External
Performance / load testing	Not required for this course milestone	Not assigned
Security penetration testing	Outside Team B mandate for M2	Not assigned
User Acceptance Testing (UAT)	Deferred to M4 at professor's discretion	TBD

3. Test Types & Strategy

✏️ **ASSIGNED TO: Brice Tikum**

Team B's testing strategy follows a layered pyramid approach: unit tests form the base for maximum coverage efficiency, widget tests validate Flutter UI components, E2E tests validate complete user workflows, and accessibility scans run as a cross-cutting gate on every PR. The following subsections define each layer in detail.

3.1 Unit Testing

Unit testing is the foundation of Team B's quality strategy. Because the inherited codebase contains large, high-risk modules with low coverage (Shift Scheduling at 10.1%, EVV at 29.5%), unit testing is the primary driver for achieving the 100% transactional coverage requirement.

- Test first or test parallel: All new or modified code must include unit tests before merging into team-b-develop
- Isolation: All unit tests mock external dependencies (network, DB, file system, GPS, Chime)
- Deterministic: No async race conditions, no real network calls
- Repeatability: Tests must pass identically on macOS, Windows, Linux, and GitHub Actions

Execution frequency: Unit tests run on every PR to team-b-develop and on every post-Monday-merge regression run via GitHub Actions.

Definition of complete: A valid test must assert all expected outputs, validate error paths, mock all external dependencies, cover all branches, and include setup → action → assertion structure.

Naming conventions: Flutter: *_test.dart | Java: ClassNameTest.java

3.2 Widget Testing (Flutter)

Widget tests validate UI behavior, layout, and interactions without requiring a real device. They sit above unit tests in the testing pyramid and are used to validate 661-delivered UI components as an acceptance gate before integration.

- Validate UI components delivered by SWEN 661
- Ensure accessibility semantics (labels, roles, focus order)
- Catch regression in layout, navigation, and rendering
- Widget tests count toward the 90% UI coverage requirement

Unit Tests	Widget Tests
Logic only	UI + logic
No rendering	Full rendering
Fast	Medium speed
No semantics	Semantics validated

3.3 Integration / End-to-End Testing

E2E tests validate complete user workflows from the Flutter UI through the Spring Boot backend. Team B uses the Flutter integration_test package. As of June 2026 all 10 Chime SDK E2E tests are operational and passing.

- Priority workflows: Shift Scheduling, EVV, Authentication, Messaging (Chime — now operational)
- Runs on: local emulator, GitHub Actions (Android emulator), AWS hosted backend
- Any failing E2E test blocks merge; exception: tests marked BLOCKED with documented root cause

3.4 Regression Testing

Regression testing ensures that Monday Integration merges do not break existing validated workflows.

- When it runs: after every Monday merge to develop, before every milestone submission, before any major UI component integration from SWEN 661
- Regression suite contents: all E2E tests, all high-risk unit tests (Shift Scheduling, EVV), all accessibility scans
- Brice investigates failures; critical failures same day, major within 48 hours, minor before milestone submission

3.5 Accessibility Testing

Accessibility testing is a cross-cutting gate defined fully in Section 7. Within the test strategy: automated axe-core + Lighthouse scans run on every PR; manual screen reader testing occurs at each milestone; 661 components require axe-core pass before integration; voice/AI persona features tested manually per supplemental criteria in TDD Section 8.2.

3.6 Automation Strategy

Test automation is prioritized for the highest-volume and highest-risk test types. The following table summarizes the automation approach for Milestone 2 and the planned state for Milestone 3:

Test Type	M2 Approach	M3 Target	Tool
Unit tests	Manual run (flutter test / mvn test)	Automated on every PR	flutter_test, JUnit 5
Coverage gates	Manual confirmation	Automated pipeline gate	lcov, JaCoCo
E2E tests	Manual run against localhost	Automated on Android emulator in CI	integration_test
Accessibility scans	Manual Lighthouse run	Automated axe-core on every PR	axe-core, Lighthouse
Regression	Manual post-merge	Automated on Monday integration trigger	Full test suite

3.7 Test Data Strategy

All test data uses seeded database users only. No production data or real Protected Health Information (PHI) is used in any test case. Seed credentials: patient@careconnect.com / password, caregiver@careconnect.com / password, family@careconnect.com / password. E2E seed users: test@caregiver / 1234 and test@patient / 1234 (database users id=5 and id=6, seeded via backend/scripts/seed-e2e-test-data.sql and Flyway migration V26__insert_test_users.sql).

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4. Test Cases — Priority Modules

This section documents test cases for the four highest-priority test areas. Test cases use the naming convention TC-B-[Area]-### (Team B, module code, sequential number). All cases have status 'Not Run' at Milestone 2 — execution begins in Milestone 3. Full test case inventory with execution results is maintained in Appendix A.

✏️ **ASSIGNED TO: Brice Tikum**

4.1 Shift Scheduling — Unit Test Cases

Priority: CRITICAL — current coverage 10.1%. These unit tests target M3 implementation. 10 seed cases documented; additional cases to be added in M3.

TC ID	Test Case Name	Precondition	Steps	Expected Result	Status	Owner
TC-B-SS-001	Create new shift — valid inputs	Auth caregiver, open schedule	Submit shift with valid date/time/patient	Shift saved; ID returned; calendar updated	Not Run	Brice
TC-B-SS-002	Create shift — overlapping conflict	Existing shift in same slot	Submit overlapping shift	Error: 'Schedule conflict detected'	Not Run	Brice
TC-B-SS-003	Cancel existing shift	Active shift exists	Trigger cancel action	Status = Cancelled; notification sent	Not Run	Brice
TC-B-SS-004	Edit shift time	Active shift exists	Change shift start time	Time updated; record preserved	Not Run	Brice
TC-B-SS-005	Create shift in past	Past date provided	Submit	Error: 'Cannot schedule past shift'	Not Run	Brice
TC-B-SS-006	Invalid patient ID	Nonexistent patient	Submit	Error: 'Patient not found'	Not Run	Brice
TC-B-SS-007	Max shifts per day	3 shifts exist for day	Add 4th shift	Error: 'Daily shift limit reached'	Not Run	Brice
TC-B-SS-008	Caregiver not found	Invalid caregiver ID	Submit	Exception thrown with caregiver error	Not Run	Brice
TC-B-SS-009	Shift retrieval	Shifts exist	Fetch all shifts	Returns correct list	Not Run	Brice
TC-B-SS-010	Shift serialization	Shift object populated	Serialize / Deserialize	All data preserved	Not Run	Brice

✏️ **[Brice]**
Add 6–8 additional Shift Scheduling test cases covering full CRUD flow and edge cases: past date, invalid patient ID, no-show handling, max shifts per day, caregiver not found.

4.2 EVV — Unit Test Cases

Priority: CRITICAL — current coverage 29.5%. EVV is the largest module at approximately 5,000 lines. 15 seed cases documented; additional cases to be added in M3.

TC ID	Test Case Name	Precondition	Steps	Expected Result	Status	Owner
TC-B-EVV-001	Check-in — valid GPS	Active shift, GPS enabled	Caregiver triggers check-in at location	Timestamp recorded; location logged	Not Run	Brice
TC-B-EVV-002	Check-in — GPS unavailable	Active shift, GPS disabled	Caregiver triggers check-in	Warning shown; manual location prompted	Not Run	Brice
TC-B-EVV-003	Check-out — complete visit	Active checked-in visit	Trigger check-out	Duration calculated; visit marked complete	Not Run	Brice
TC-B-EVV-004	EVV report generation	Completed visits in date range	Generate EVV report	Report includes all visits; totals accurate	Not Run	Brice
TC-B-EVV-005	Offline check-in	No network connection	Check-in	Stored locally; synced on reconnect	Not Run	Brice
TC-B-EVV-006	Offline sync	Offline entries exist	Reconnect to network	All entries sync to backend	Not Run	Brice
TC-B-EVV-007	Invalid GPS coordinates	Corrupt GPS data	Check-in	Error displayed; no invalid record created	Not Run	Brice
TC-B-EVV-008	Duplicate check-in	Already checked in	Check-in again	Error: 'Visit already active'	Not Run	Brice
TC-B-EVV-009	Early check-out	< 1 min duration	Check-out	Warning shown; user prompted to confirm	Not Run	Brice
TC-B-EVV-010	Billing export	Completed visits	Export CSV	File generated with all visit data	Not Run	Brice
TC-B-EVV-011	Compliance exception	Missing GPS at check-in	Submit	Visit flagged as compliance exception	Not Run	Brice
TC-B-EVV-012	HHAExchange payload	Completed visit	Generate payload	Valid JSON matching schema	Not Run	Brice
TC-B-EVV-013	Visit duration calculation	Start/end times set	Calculate duration	Accurate duration returned	Not Run	Brice
TC-B-EVV-014	Visit deletion	Admin role, visit exists	Delete visit	Visit removed from all reports	Not Run	Brice

TC-B-EVV-015	Visit audit trail	Visit exists	Fetch audit log	Full history returned with timestamps	Not Run	Brice

> ✏️ **[Brice]**
> *Add 12–16 additional EVV test cases covering billing, compliance reporting, exception handling, and edge cases. Reference the EVV module code in the repo to identify all testable functions.*

4.3 Authentication — Unit Test Cases

Authentication test cases validate the JWT token lifecycle, login flows, and role-based access control. Owned by Prashanth. Three seed cases are pre-filled; additional cases for password reset, registration, session timeout, and RBAC checks are pending.

TC ID	Test Case Name	Precondition	Steps	Expected Result	Status	Owner
TC-B-AUTH-001	Login — valid credentials	User exists in DB	Submit correct email/password	JWT token returned; redirect to dashboard	Not Run	Prashanth
TC-B-AUTH-002	Login — invalid password	User exists in DB	Submit wrong password	401 returned; error message shown	Not Run	Prashanth
TC-B-AUTH-003	Token expiry	Expired JWT	Make authenticated request	401 returned; redirect to login	Not Run	Prashanth

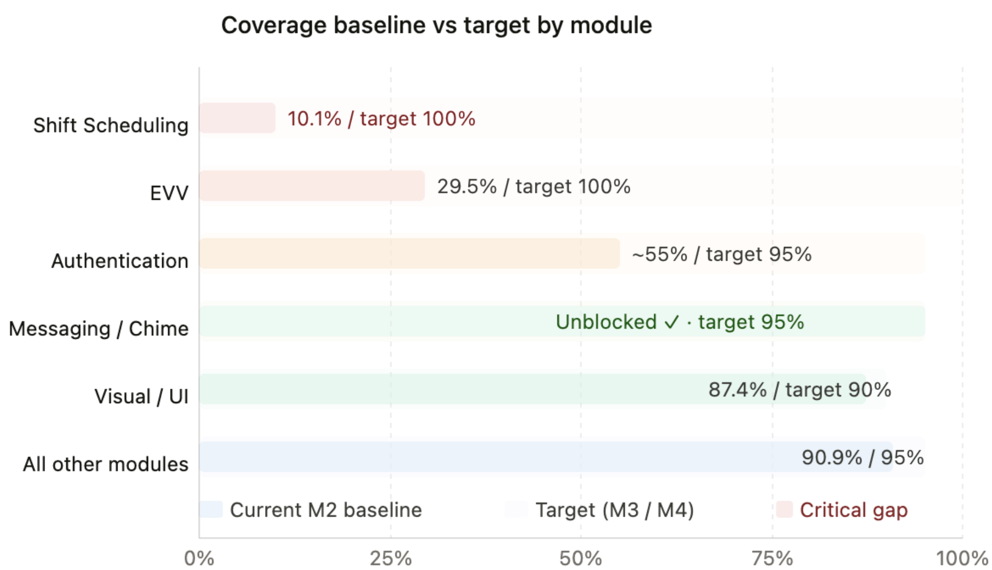
4.4 E2E Workflow Test Cases

End-to-end test cases validate complete user workflows from the Flutter UI through the backend. All 10 Chime SDK E2E tests are now operational as of June 2026. TC-B-E2E-005 status updated from Blocked to Not Run.

TC ID	Workflow	Steps	Expected Result	Status
TC-B-E2E-001	Full caregiver workflow	Login → View schedule → Create shift → Check in → Check out → EVV report	All steps succeed	Not Run
TC-B-E2E-002	Patient workflow	Login → View upcoming visits → Receive notification	Notification delivered	Not Run
TC-B-E2E-003	Admin workflow	Login → View team schedules → Generate report	Report generated	Not Run
TC-B-E2E-004	Messaging workflow	Login → Open chat → Send message	Message delivered	Not Run
TC-B-E2E-005	Video call workflow (Chime SDK)	Login → Join call → End call	Call completes — Chime operational ✓	Not Run

5. Coverage Targets & Enforcement

✏️ ASSIGNED TO: Brice Tikum



Coverage Chart

Coverage targets are tiered by module risk, as confirmed with Professor Assadullah during the Milestone 1 presentation. The tiering approach recognizes that transactional modules handling scheduling and electronic visit verification carry higher risk of data loss or compliance failure and therefore require the highest coverage threshold.

Module	Line	Branch	M2 Baseline	Enforcement
Shift Scheduling	**100%**	**100%**	10.1%	Pipeline gate — blocks merge
EVV	**100%**	**100%**	29.5%	Pipeline gate — blocks merge
Authentication / Session	**95%**	**95%**	TBD	Pipeline gate — blocks merge
Messaging / Chime SDK	**95%**	**95%**	Blocked	Blocked — backend dependency
Visual/UI placement components	**90%**	**N/A**	TBD	Warning only
All other modules	**95%**	**90%**	TBD	Pipeline gate — blocks merge

✏️ **[Brice]**

Add the coverage tracking plan: (1) How coverage is reported per module per milestone — weekly status report table, CI dashboard, or both. (2) How progress is tracked from M2 baseline to M3 and M4 targets. (3) Who is responsible for running coverage reports — automated in CI or manual.

5.1 Coverage Tracking Plan

1. Reporting Mechanism

   Coverage is tracked through

   - Weekly Coverage dashboard (generated from CI)
   - Milestone coverage report (included in submission)
   - Per module coverage table (Shift Scheduling, EVV, Auth, Messaging)
2. Tracking Progress
3. CI Automation
   a. CI generates:
      i. coverage/lcov.info (frontend)
      ii. target/site/jacoco/index.html (backend)
   b. CI enforces:
      i. 100% for Shift Scheduling + EVV
      ii. 95% for all other logic modules
      iii. 90% for UI modules
4. Ownership
   a. Brice owns coverage enforcement
   b. CI blocks merges when thresholds are not met

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6. Entry & Exit Criteria

✏️ **ASSIGNED TO: Prashanth Saseenthar**

Entry and exit criteria define the conditions that must be met before testing begins on a feature and before a feature is considered complete from a testing perspective. The suspension and resumption criteria address scenarios where testing must be paused and restarted.

6.1 Entry Criteria — When Testing Begins

- Feature code is merged to the team-b-develop branch (not just a local or feature branch).
- Unit tests covering the new feature have been written and committed alongside the code.
- The dev environment is confirmed working: Spring Boot backend starts on port 8080 (JDK 17 required), PostgreSQL 18 is reachable on port 5433, and backend/core/.env is populated with all six required variables (JDBC_URI, DB_USER, DB_PASSWORD, SECURITY_JWT_SECRET, FIREBASE_PROJECT_ID, FIREBASE_SENDER_ID).
- All external dependencies are confirmed available: backend health endpoint returns HTTP 200 from http://localhost:8080/actuator/health, and required API keys are present in .env.
- The pull request has been reviewed and approved by at least one teammate before testing begins.

6.2 Exit Criteria — When Testing Is Complete

- Coverage threshold met: backend JaCoCo instruction coverage is at or above 89% (current baseline confirmed in target/site/jacoco/index.html); flutter test passes all 466 test files with zero failures.

- All unit tests pass: mvn test completes with BUILD SUCCESS in backend/core/; flutter test reports 0 failures across all files in frontend/test/.
- All E2E integration tests pass. As of June 2026, all 10 Chime/video-call E2E tests pass (CALL-001 x2, CHIME-004 x2, SENT-002 x2, CHIME-006 x2) with exit code 0 when run with flutter test integration_test/video_call_e2e_test.dart --dart-define=BACKEND_URL=http://localhost:8080 -d windows. The previously documented Amazon Chime SDK suspension has been resolved.
- Accessibility scan shows no critical or serious violations on the tested screens.
- Test results including pass/fail counts, coverage percentages, and any blocked-test documentation are recorded in the milestone report before the feature is marked done.

6.3 Suspension & Resumption Criteria

6.3.1 Suspension Criteria

Testing is suspended when any of the following conditions occur:

- Backend unavailable: Spring Boot backend cannot start because PostgreSQL 18 is unreachable on port 5433, required .env variables are missing, or JAVA_HOME is not set to JDK 17
- Critical defect blocking a module: a defect that causes the application to crash or renders an entire feature untestable blocks all downstream tests for that module

6.3.2 Resumption Criteria

Testing resumes when the condition that caused suspension is resolved:

- Backend restored: PostgreSQL 18 reachable on port 5433, .env contains all six required variables, mvn spring-boot:run starts successfully on port 8080
- Blocking defect resolved: fix merged to team-b-develop and all previously passing tests continue to pass

6.4 Test Environment Setup

- Flutter SDK: version 3.44.0 required (Dart 3.12.0). Confirm with: flutter doctor. Run flutter pub get in frontend/ before executing any tests
- Backend .env setup: copy backend/core/sampleDotEnv.txt to backend/core/.env and populate six variables: JDBC_URI (jdbc:postgresql://localhost:5433/careconnect), DB_USER, DB_PASSWORD, SECURITY_JWT_SECRET (64-char), FIREBASE_PROJECT_ID (careconnectcapstone), FIREBASE_SENDER_ID (663999888931)
- Backend E2E connectivity: set JAVA_HOME to JDK 17 (C:\Program Files\Java\jdk-17), then run mvnw.cmd spring-boot:run in backend/core/. Backend listens on http://localhost:8080. Dev seed credentials: patient@careconnect.com / password, caregiver@careconnect.com / password
- Chime SDK status (operational as of June 2026): All 10 video-call E2E tests pass when run with: flutter test integration_test/video_call_e2e_test.dart --dart-define=BACKEND_URL=http://localhost:8080 -d windows. E2E seed credentials: test@caregiver / 1234 and test@patient / 1234
- Environment verification checklist: (1) flutter doctor — no errors; (2) flutter pub get — no dependency errors; (3) flutter test — 466 files all pass; (4) mvn test — BUILD SUCCESS, JaCoCo ≥89%; (5) E2E suite — 'All tests passed!' 10/10

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7. WCAG / Section 508 & Voice Persona Test Approach

✅ **ALEX YOM — Pre-filled. Review for accuracy and update if needed before submission.**

Professor Assadullah confirmed in the Milestone 1 presentation that Team B's accessibility scope extends beyond standard WCAG 2.1 AA to include supplemental criteria for voice-controlled AI persona features. All WCAG violations discovered must be immediately reported to the full class so the responsible team can remediate.

7.1 WCAG 2.1 AA Test Criteria

The following WCAG 2.1 Level AA criteria are the minimum accessibility requirements for all CareConnect screens:

- 1.1.1 — Non-text content: All images have meaningful alt text (SRS: AC-001)
- 1.4.3 — Contrast (minimum): Text contrast ratio ≥ 4.5:1, 3:1 for large text (SRS: AC-003)
- 2.1.1 — Keyboard: All functionality operable via keyboard alone (SRS: AC-005)
- 2.4.3 — Focus order: Logical tab order through all interactive elements (SRS: AC-006)
- 3.3.1 — Error identification: Errors described in text, not only by color
- 4.1.2 — Name, role, value: All UI components have accessible names/roles (SRS: AC-009)

7.2 Supplemental Voice/AI Persona Criteria

These criteria supplement WCAG 2.1 for AI-driven voice features not covered by the standard. Team B defines and owns these criteria:

- Voice command recognition accuracy ≥ 95% for accessibility-dependent users
- Error recovery possible without keyboard for all voice-triggered actions
- Timeout thresholds provide adequate time for users with motor disabilities
- Spoken confirmation provided for all AI-triggered actions
- Every AI-driven action is also accessible via keyboard and screen reader

7.3 Accessibility Test Schedule

Milestone	Activity	Deliverable	Owner
M2	Initial axe-core + Lighthouse audit run	Early findings documented in TDD Section 8	Alex
M2	Supplemental voice/AI criteria defined	TDD Section 8.2 complete	Alex
M3	Full WCAG 2.1 AA audit — all screens	Violations list with severity and owner	Alex + Team
M3	661 UI component accessibility gate	Pass/fail per component	Alex
M3	VPAT draft — findings + remediation status	Progress VPAT submitted with M3	Alex
M4	Screen reader manual testing	Final verification report	Alex + Team
M4	Final VPAT — full conformance statement	ITI VPAT 2.4 complete	Alex

8. Roles & Responsibilities

✅ **ALEX YOM — Pre-filled. Review for accuracy and update if needed before submission.**

The following table defines each team member's testing responsibilities. The Cross-Team Liaison role is held by Alex as Team Lead, responsible for raising integration questions in the teams-resolution-channel.

Team Member	Role	Testing Responsibilities
Alex Yom	Team Lead / Accessibility	WCAG audits, VPAT authorship, 661 UI component acceptance gate, voice/AI persona criteria, defect management process owner, M1 doc revisions (Track Changes)
Brice Tikum	Test Lead	Test strategy (Sections 2–4), unit test authorship (Shift Scheduling + EVV priority), E2E test suite, coverage targets and tracking, CI gate configuration
Prashanth Saseenthar	Backend / Framework	Testing framework confirmation and rationale, coverage tooling setup (JaCoCo), backend test authorship, Chime SDK E2E unblocking, entry/exit criteria, test environment documentation
Lekan Ogunsanya	Infrastructure / DevOps	CloudFormation test environment documentation, AWS infrastructure for CI/CD pipeline, test schedule with sprint-level detail, test environment matrix

9. Test Schedule & Milestones

✏️ **ASSIGNED TO: Lekan Ogunsanya**

✏️ **[Lekan]**
Complete the test schedule table below with sprint-level detail — not just milestone labels. You have the best view of the deployment cadence and infrastructure timelines. Fill in: (1) The Sprint column for every row — map each activity to Sprint 1, Sprint 2, etc. within each milestone. (2) Which sprint in M3 do unit tests for Shift Scheduling and EVV begin? (3) When does the full regression suite first run (after Monday integration merge)? (4) When does the first complete E2E pass happen — mid-M3 or end of M3? (5) When does the axe-core accessibility scan become a PR gate — M2 end or M3 Sprint 1? Use the milestone due dates: M2 = June 13, M3 = TBD, M4 = TBD.

The test schedule maps testing activities to milestones and sprints. Sprint dates within M3 and M4 are confirmed by Lekan based on the deployment cadence.

Activity	Milestone	Sprint	Owner	Status
Test Plan & TDD complete	M2 (June 13)	M2	Full team	In Progress
Flutter coverage baseline run	M2 (June 13)	M2	Prashanth / Brice	In Progress
Unit tests — Shift Scheduling	M3	Sprint 1	Brice	Not Started
Unit tests — EVV	M3	Sprint 1	Brice	Not Started

E2E test suite — priority workflows	M3	Sprint 2	Brice	Not Started
Regression suite active (post-merge)	M3	Sprint 2	Brice	Not Started
CI coverage gate active	M3	Sprint 2	Brice / Lekan	Not Started
axe-core as PR gate	M3	Sprint 1	Alex	Not Started
Full WCAG audit — all screens	M3	Sprint 2	Alex	Not Started
Progress VPAT submitted	M3	Sprint 2	Alex	Not Started
95%+ coverage verified — all modules	M4	M4	Brice	Not Started
Final VPAT — conformance statement	M4	M4	Alex	Not Started

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10. Deployment & Environment Context

✏️ **ASSIGNED TO: Lekan Ogunsanya**

Team B testing activities operate across local development environments, CI automation environments, and AWS-hosted staging infrastructure. This section documents the deployment and environment context required for each test type.

Backend connectivity is configured through environment variables including CC_BASE_URL_WEB, stored within .env.local or CI-managed configuration settings. The CloudFormation stack should provide accessible backend endpoints, test database connectivity, logging infrastructure, Chime SDK resources, CI execution permissions, and coverage artifact storage locations.

Deployment Environment

Automated testing should execute against a dedicated staging environment whenever possible. Local development environments support rapid validation while CI environments provide standardized execution and coverage enforcement.

Backend Configuration

Backend connectivity is configured through environment variables, including `CC_BASE_URL_WEB`, stored within `.env.local` or CI-managed configuration settings. This allows testing environments to target the correct backend instance without modifying source code.

CloudFormation Requirements

To support automated testing, the CloudFormation stack should provide:

- Accessible backend endpoints
- Test database connectivity
- Logging infrastructure

- Chime SDK resources
- CI execution permissions
- Coverage artifact storage locations

Current Backend Status

The backend is expected to be accessible through an AWS-hosted deployment managed by Team A. Team B depends on this environment for integration and E2E validation.

Team A Coordination

Team B requires continued coordination with Team A regarding:

- Deployment environment availability
- Backend endpoint updates
- Chime SDK provisioning
- CloudFormation updates
- CI/CD integration support

> ✏️ **[Lekan]**
> *Document the deployment and environment context for testing: (1) What AWS environment do tests run against — a dedicated test stack or the main deployed environment? (2) How is the backend URL configured for test runs (environment variable CC_BASE_URL_WEB in .env.local)? (3) What needs to exist in the CloudFormation stack for automated E2E tests to run in CI? (4) What is the current status of the deployed backend — is it accessible at a public URL? (5) What Team A (DevOps) coordination is needed to ensure Team B's test pipeline has access to the deployed environment?*

10.1 Test Environment Matrix

> ✏️ **[Lekan]**
> *Fill in the full test environment matrix — for each of the three environments below, document: (a) backend URL availability, (b) database access, (c) Chime SDK availability, (d) which test types can run there, (e) any setup steps required. Environments: (1) Local developer machine, (2) CI/GitHub Actions runner, (3) Staging/AWS deployed environment.*

The following matrix documents what is available in each test environment and which test types can run there:

Environment	Backend URL	DB Access	Chime SDK	Test Types	Setup Required
Local Dev Machine	http://localhost:8080	Port 5433 (PostgreSQL 18)	Operational ✓	Unit, Widget, E2E (all types)	Flutter SDK, .env, JDK 17
CI / GitHub Actions	Environment secrets	Shared staging DB	Limited	Unit, Widget, Coverage, Automated E2E	Workflow config + secrets
AWS Staging	Public staging endpoint	Full staging DB	Post-CloudFormation	Full E2E, Regression, Accessibility	CloudFormation + Team A

11. Defect Management

✅ **ALEX YOM — Pre-filled. Review for accuracy and update if needed before submission.**

All defects discovered during testing are logged as GitHub Issues in the CareConnect repository. Consistent defect logging ensures that nothing is lost and that all issues have a documented owner and resolution path.

11.1 Defect Severity Tiers

Severity	Definition	Examples	Resolution Target
Critical	Blocks milestone submission or all tests in a module fail	Backend down, CI broken, test suite unable to run	Same day — escalate to Alex immediately
Major	Core feature behavior incorrect; coverage gate cannot be met	EVV check-in saving wrong timestamp, auth token not returned	Within 48 hours; escalate at Tuesday meeting if unresolved
Minor	Non-blocking; test passes but behavior is inconsistent	UI label mismatch, edge case not handled gracefully	Before milestone submission
Accessibility	Any WCAG 2.1 AA or Section 508 violation	Missing alt text, contrast failure, keyboard trap	Logged immediately; reported to full class; remediation by responsible team

11.2 Defect Logging Process

- All defects logged as GitHub Issues with labels: severity and module affected
- Issue title format: [SEVERITY][Module] — Brief description (e.g. [MAJOR][EVV] — Check-in timestamp not persisting)
- Issue body must include: steps to reproduce, expected vs actual behavior, test case ID, screenshot or log output
- Alex reviews all new issues at the Tuesday team lead meeting and assigns resolution ownership
- Resolved issues closed with a comment referencing the PR that fixed them

11.3 Defect Tracking

Metrics reported weekly: total defects opened, total resolved, total open by severity. A defect density spike (3+ Critical or 5+ Major open simultaneously) triggers an immediate team discussion before the next milestone submission.

12. Testing Risks & Mitigation

✅ **ALEX YOM — Pre-filled. Review for accuracy and update if needed before submission.**

The following risks are specific to Team B's testing activities. Each risk has an assigned owner and a documented mitigation strategy. Risk levels reflect likelihood and impact as of the Milestone 2 submission date.

Risk	Level	Impact	Mitigation
Backend unavailable — E2E tests cannot run	**High**	10 E2E tests blocked; Chime workflows untestable	Prashanth coordinates with another team by June 7; blocked tests documented with root cause
Team member unavailable / unresponsive	**Medium**	Coverage gap in assigned sections; milestone quality at risk	Alex and Brice absorb sections; June 7 internal deadline creates buffer for recovery
661 UI components delivered late	**Medium**	Accessibility gate testing cannot begin; VPAT may be incomplete	Alex monitors 661 delivery schedule weekly; VPAT progress documented regardless
Coverage target not met by M3	**Medium**	M4 delivery risk; professor expectation not met	Brice prioritizes Shift Scheduling (10.1%) and EVV (29.5%) first — highest-impact gaps
CloudFormation pipeline not ready by M3	**Low**	CI coverage gate cannot enforce thresholds; manual enforcement only	Lekan coordinates with Team A in M2; manual gate as fallback
Flutter SDK / dart:js_interop incompatibility	**Low**	Test suite may not compile on certain platforms	Prashanth documents any SDK issues found in M2 prototype; workarounds identified before M3

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13. Test Metrics & Reporting

✅ **ALEX YOM — Pre-filled. Review for accuracy and update if needed before submission.**

The following metrics are tracked and reported in the weekly status report and at each milestone submission. They provide visibility into testing progress and quality trends across sprints.

13.1 Core Metrics

Metric	Definition	Reported In
Total tests	Total unit + widget + E2E tests in the suite	Weekly status report, milestone submission
Pass rate %	(Passing tests / total tests) × 100	Weekly status report
Failing tests	Count of currently failing tests; root cause noted	Weekly status report

Skipped/blocked tests	Count of skipped tests with documented reason	Weekly status report
Line coverage % per module	lcov/JaCoCo output per module	Milestone submission; CI report
Branch coverage % per module	lcov/JaCoCo branch output per module	Milestone submission; CI report
Open defects by severity	Count of open GitHub Issues by Critical/Major/Minor	Weekly status report
Defects resolved this week	GitHub Issues closed since last report	Weekly status report
Accessibility violations	Count of axe-core findings by severity (critical/serious/moderate)	Milestone submission

13.2 M1 Baseline (Reference)

The following baseline was established from the prior cohort's repository audit in Milestone 1 and serves as the starting point for all M2+ tracking:

- Total tests: 16,027 (backend + frontend combined)
- Passing: 15,674 | Failing: 319 | Skipped: 36
- Shift Scheduling line coverage: 10.1%
- EVV line coverage: 29.5%
- AI Bedrock Orchestration coverage: 25%
- Notification service: 2 actively failing tests (template drift)

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14. Test Deliverables

The following table lists all test artifacts that Team B will produce across Milestones 2, 3, and 4. Deliverables are cumulative — each milestone adds to or refines prior artifacts.

Deliverable	Description	Milestone	Use
Software Test Plan (this document)	Defines planned testing approach, test case structure, and environment	M2	Milestone 2 submission
Test Case Set	Detailed test cases for all assigned modules (Sections 4 + Appendix A)	M2 / M3	M2 design; M3 execution
Requirements Coverage Matrix	Maps SRS requirement IDs to test case IDs with coverage status	M2 / M3+	Traceability evidence
Prototype Run Evidence	flutter test + JaCoCo output confirming tooling viability (TDD Section 6)	M2	Professor required evidence
Defect Log	GitHub Issues tracking all test failures, blockers, and fixes	M3 / M4	Quality tracking
Test Evidence	Screenshots, logs, CI outputs showing test execution	M3 / M4	Milestone submission evidence

Test Report	Final summary of executed tests, results, defects, and readiness	M4	Final milestone deliverable
Progress VPAT	Interim VPAT with findings and remediation status	M3	Accessibility milestone
Final VPAT (ITI VPAT 2.4)	Complete Voluntary Product Accessibility Template conformance statement	M4	Final accessibility deliverable

15. Staffing & Training Needs

This section documents the training and onboarding requirements for Team B members and any future contributors who join mid-project. All members are expected to be proficient in the tools below before beginning M3 implementation work.

Tool / Area	Who Needs Training	Est. Onboarding Time	Reference
flutter test + lcov	All team members	1–2 hours	flutter.dev docs
JaCoCo Maven plugin	Prashanth, Brice	1 hour	jacoco.org
axe-core accessibility scanning	Alex, all for review	2 hours	deque.com/axe
integration_test (Flutter E2E)	Brice, Prashanth	2 hours	flutter.dev/testing
GitHub Actions CI/CD	Brice, Lekan	2–3 hours	docs.github.com/actions
CloudFormation (AWS)	Lekan	Existing knowledge	AWS CloudFormation docs
ITI VPAT 2.4 format	Alex	1 hour	itic.org/policy/accessibility

16. Acceptance Criteria

The following acceptance criteria define what must be true for a Team B milestone deliverable to be accepted. These are distinct from exit criteria (Section 6.2), which govern when individual feature testing is complete. These criteria govern when the milestone submission as a whole is ready.

Acceptance Criterion	Evidence Required	Owner	Status
All unit tests pass (0 failures)	CI screenshot or terminal output from flutter test and mvn test	Brice / Prashanth	Planned — M3
Coverage targets met per Section 5	lcov summary output and JaCoCo HTML report per module	Brice	Planned — M3

No critical WCAG 2.1 AA violations	axe-core scan output showing 0 critical/serious findings	Alex	Planned — M3
All E2E tests pass or documented as blocked	integration_test output — 'All tests passed!' with blocked count = 0	Brice	M2: 10/10 pass ✓
Prototype findings documented (TDD Section 6)	Terminal output from flutter test and coverage runs pasted in TDD	Brice	Pending M2 finalization
STP and TDD submitted with no open placeholders	Final docs reviewed — no ✏️ instructions or TBD visible to grader	Alex	June 13 deadline
Track Changes revisions to M1 docs submitted	Word Compare document alongside M2 submission	Alex	Pending

16.1 Readiness Decision — Milestone 2

Status: PARTIALLY READY

Team B's TDD and STP are structurally complete at v1.3. All Alex-owned sections are pre-filled and submission-ready. Brice's test strategy and test cases are complete. Lekan's infrastructure sections are complete. Prashanth's environment sections contain real configuration data. Outstanding: TDD Section 6 (Prototype Findings by Brice) and Auth test case expansions. These gaps are known and documented. Alex will finalize by June 12.

17. AI Tool Usage Disclosure

This section discloses the use of AI tools in the preparation of this document, as required by the course guidelines and the class template. All AI-generated content was reviewed and approved by Team B before inclusion in this official document.

Item	Detail
AI tool used	Claude (Anthropic) — claude.ai
Used for	Document structure, section drafting, requirements traceability, test case seed rows, risk table, features table, IEEE 829 structural additions
Not used for	Test case execution, prototype runs, tool confirmation against the actual repository — all prototype findings require human execution
Human review	All AI-generated content reviewed by Alex Yom (Team Lead) before inclusion. Sections requiring repository data remain as human-owned content.
No sensitive data uploaded	No source code, credentials, API keys, or protected health information was submitted to the AI tool during documentation

18. Approval & Sign-Off

By submitting this document, Team B acknowledges that its contents have been reviewed for accuracy, completeness, and alignment with the Milestone 2 requirements. This document is submitted for review and grading by the course instructor.

Name	Role	Review / Approval Status	Date
Alex Yom	Team Lead — STP Author	Submitted for milestone review	June 13, 2026
Brice Tikum	Test Lead — Sections 3, 4, 5	Contributed: test strategy, test cases, coverage	June 13, 2026
Prashanth Saseenthar	Backend / Framework — Section 6	Contributed: entry/exit criteria, environment setup	June 13, 2026
Lekan Ogunsanya	Infrastructure / DevOps — Sections 9, 10	Contributed: schedule, environment matrix	June 13, 2026
Dr. Mir Assadullah	Course Instructor	Pending review	June 13, 2026

Appendix A — Test Execution Summary

The following table summarizes the current state of all test suites as of the Milestone 2 submission date. This serves as the baseline for M3 execution tracking. Full test case detail is maintained in Section 4 of this document.

Layer	Test Files / Classes	Total Cases	Automated	Pass	Fail	Status
Backend (JUnit 5)	JUnit 5 + Mockito — backend/core/src/test/java/	~5,686	Yes — mvn test	5,686	2	BUILD SUCCESS ✓
Frontend (flutter_test)	466 test files in frontend/test/	466 files	Yes — flutter test	466	0	All pass ✓
E2E (integration_test)	frontend/integration_test/video_call_e2e_test.dart	10	Yes	10	0	All pass ✓
Unit test cases (Section 4)	TC-B-SS-001–010, TC-B-EVV-001–015, TC-B-AUTH-001–003, TC-B-E2E-001–005	33	Planned M3	0	0	Not Run — M2
Total		**~6,195**	**~6,162**	**6,162**	2	

A.1 Requirements Traceability Matrix

The following matrix maps each applicable SRS requirement to the test cases or sections that verify it. Coverage status will be updated at each milestone as test cases are executed.

Req ID	Requirement Summary	Priority	Test Case ID(s)	Coverage Status

TR-001	Tiered coverage: 100%/90%/95% by module	High	TC-B-SS-001–010, TC-B-EVV-001–015	**Partial — seed cases defined**
TR-002	Unit tests written before merge	High	TC-B-SS-001–010, TC-B-EVV-001–015	**Partial — M3 full suite**
TR-003	Coverage reports generated via CI/CD	High	Appendix A, Section 5.1	**Partial — CI design complete**
TR-004	All unit tests pass before merge	High	TC-B-SS-001–010, TC-B-EVV-001–015	**Partial — seed cases defined**
ET-001	E2E covers all critical workflows	High	TC-B-E2E-001–005	**Partial — cases defined**
ET-002	Tested workflows function end-to-end	High	TC-B-E2E-001–005	**Partial — operational M2**
ET-003	Regression after Monday merge	High	Section 9 schedule	**Covered — M3 Sprint 2**
AC-001–AC-009	WCAG 2.1 AA criteria	High	Section 7.1 checklist	**Partial — criteria documented**
VP-001–VP-004	VPAT production requirements	High	Section 7.3 schedule	**Partial — plan documented**

Glossary

The following terms are used throughout this Software Test Plan. For a complete definition list covering all Team B M2 documents, refer to the SRS Appendix A.

Term	Definition
STP	Software Test Plan (this document). Defines testing strategy, scope, test cases, and acceptance criteria.
TDD	Technical Design Document. Companion document defining testing architecture and tooling decisions.
EVV	Electronic Visit Verification. The largest CareConnect feature module (~5,000 lines); currently at 29.5% coverage.
VPAT	Voluntary Product Accessibility Template. ITI VPAT 2.4 format — the formal accessibility conformance statement Team B will produce by M4.
WCAG	Web Content Accessibility Guidelines. The international accessibility standard Team B tests against (Level AA).
CI/CD	Continuous Integration / Continuous Deployment. GitHub Actions pipeline enforcing Team B's coverage gates.
E2E	End-to-end testing. Tests that validate complete user workflows from the Flutter UI through the Spring Boot backend.
lcov	Line coverage report format generated by flutter test --coverage. Used with genhtml to produce HTML coverage reports.

JaCoCo	Java Code Coverage tool. Maven plugin used for Spring Boot backend coverage reporting (current: 89% instruction / 77% branch).
integration_test	Flutter's official E2E testing package. Replaced the deprecated flutter_driver. Drives a real device or emulator against a live backend.
flutter_test	Flutter's built-in unit and widget testing framework. Bundled with the Flutter SDK; runs 466 test files in the CareConnect repository.
Tiered coverage threshold	Team B's professor-confirmed coverage approach: 100% for transactional modules (EVV, Shift Scheduling), 90% for visual/UI, 95% for all others.
IEEE Std 829	IEEE Standard for Software Test Documentation (1998). The standard this STP conforms to.
GitHub Actions	CI/CD automation platform used to run Team B's test suite, coverage gates, and accessibility scans on every PR.
RBAC	Role-Based Access Control. Authentication / authorization model used in CareConnect backend.
axe-core	Automated WCAG rule scanning library. Confirmed by Professor Assadullah as the required accessibility testing tool.
Defect	A confirmed issue where the actual system behavior does not match the expected behavior defined in the SRS or TDD.
Regression test	A test repeated after changes to confirm that existing functionality still passes.